Data Analytics Capstone

Aksanti Global Market Research (AGMR):

NLP-Based Financial Event Classification and ML Macroeconomic Forecasting

for Sub-Saharan Africa

Interim Report

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GitHub

https://github.com/ppzanguilo/AGMR_Project

Introduction

Background and Context

Angola is among the top 10 economies in Sub-Saharan Africa and Africa's top 5 oil producers, yet it lacks a widely available, open, AI-driven financial intelligence platform. Decision-makers in government, banking, and investment rely on manual tracking of financial news and traditional ARIMA-based macroeconomic projections that do not fully leverage the predictive power of modern machine learning. This research addresses that gap through two parallel tracks: First, an NLP classifier for financial market events in Portuguese and English Angolan financial news, and second, ML forecasting models for Angola's macroeconomic indicators and commodity prices that directly drive Angola's fiscal and monetary environment.

The research serves a dual purpose. Academically, it fills an empirically verified literature gap as no peer-reviewed NLP study has classified financial events from Portuguese-language African financial text, and no ML forecasting study has applied ensemble methods to Angola-specific CPI inflation, USD/AOA exchange rate, and real interest rate series as primary targets. Commercially, it provides the scientific foundation for the Aksanti Global Market Research (AGMR) database, Angola's first AI-driven open financial intelligence platform with causality analysis embedded, to be deployed in the near future.

Problem Statement

Angola lacks an automated, scalable financial intelligence system capable of classifying financial market events from Portuguese-language news and generating ML-based macroeconomic and commodity forecasts. Existing approaches rely on manual news monitoring and linear ARIMA-based projections that fail to account for nonlinear regime shifts and cross-series dependencies. No labelled Angolan financial news corpus exists in the public domain, and no peer-reviewed ML forecasting study has applied ensemble methods to Angola-specific macroeconomic series, CPI inflation, the USD/AOA exchange rate, and the real interest rate, as primary targets. This research addresses both gaps through a dual-track methodology combining NLP event classification and ML forecasting, delivering the validated scientific infrastructure required for the AGMR platform.

Purpose of the Study

The study aims to first construct and validate an NLP classifier for six financial market event categories from Angolan Portuguese and English financial text. Second, develop and benchmark XGBoost and Random Forest forecasting models against ARIMA baselines for two frequency-consistent monthly groups—a Macroeconomic Group (CPI inflation, USD/AOA exchange rate, real interest rate) and a Commodity Group (Brent crude oil, natural gas, diamond, gold). Third, examine the causal structure among and between these series using Granger causality and event-study analysis. Fourth, integrate both the classification and forecasting models into a unified, deployable AGMR database architecture.

Interim Project Status

Completed

- Macroeconomic Time Series data collection
- Commodity Time Series data collection
- EDA for the time series data

In Progress

- NLP corpus annotation: article collection from ANGOP, Jornal de Angola, Expansão, Novo Jornal, BNA press releases, Reuters Africa, Bloomberg Africa, and AllAfrica.com.
- Time-series data cleaning and stationarity pre-checks (Augmented Dickey-Fuller tests) for both macroeconomics and commodity groups.
- Exploratory data analysis (EDA): distributional analysis, structural break visualisation at six candidate dates (2002, 2008, 2014, 2016, 2020, 2022).

Pending

- ARIMA baseline estimation via AIC minimisation for all seven series.
- XGBoost and Random Forest model training on 1960–2020 data
- Structural break regime analysis using Chow tests and CUSUM instability confirmation.
- Granger causality matrices: 3×3 (macroeconomic) and 4×4 (commodity)
- Event-study impact matrix: 6×7 event-to-series cumulative abnormal return (CAR) map.
- BERTimbau fine-tuning and McNemar’s test comparison against DistilBERT and TF-IDF + SVM baseline.

Scope and Objectives

The study covers sub-Saharan Africa, with Angola as the primary market context, over 1960–2025. Six financial event categories are targeted: Crisis, IPO/Capital Market, M&A, Bankruptcy, Regulatory/Policy, and Natural Disaster. Seven time series are organised into two monthly frequency groups: a Monthly Macroeconomic Group (CPI Inflation, USD/AOA Exchange Rate, Real Interest Rate) sourced from the World Bank, and a Monthly Commodity Group (Brent crude oil, natural gas, diamond, gold) sourced from the Federal Reserve Bank of St. Louis (FRED). All series are modelled at monthly frequency, eliminating spurious autocorrelation, look-ahead bias, and inconsistent lag structures.

Four objectives drive the study. First, construct and validate an NLP event classifier achieving a significantly good and acceptable F1 according to the literature; Second, develop XGBoost and Random Forest forecasting models for both monthly groups benchmarked against ARIMA baselines; Third, examine the effect of Angola's structural breaks (2002, 2008, 2014, 2016, 2020, 2022) on model accuracy; and fourth, integrate both tracks into a unified, deployable AGMR database architecture. The latter will be reserved for future studies.

Research Questions

RQ1 (NLP): How accurately does a fine-tuned BERTimbau model classify financial market events from Angolan Portuguese-language news compared to DistilBERT and a TF-IDF + SVM baseline, and does the language-specific model produce a statistically significant improvement in macro F1 at $\alpha = 0.05$?

RQ2 (NLP): Which of the six event categories is most and least accurately classified, and what linguistic or data features explain the performance differential?

RQ3 (ML Forecasting): Do ensemble ML methods produce statistically significantly lower MAPE than an ARIMA baseline within each monthly group, and to what extent do cross-series features improve forecast accuracy?

RQ4 (ML Forecasting): Does training on the post 2014 structural break regime improve out-of-sample forecast accuracy within each group compared to full-sample models?

RQ5 (Causation): What is the full causal structure between the series? Which monthly macroeconomic series Granger-cause each other, which monthly commodity series Granger-cause each other, and which classified event categories cause statistically significant deviations in each of the seven AGMR series? RQ5 produces a 3×3 Granger matrix for the Monthly Macroeconomic Group (6 directed pairs) and a 4×4 Granger matrix for the Monthly Commodity Group (12 directed pairs), totalling 18 directed tests at monthly frequency. The event-study dimension produces a 6×7 event-to-series causal impact map across all 42 combinations.

Literature Survey

Literature Review Approach

The articles were screened in Google Scholar and SSRN using different keywords, NLP financial event classification, BERT financial text, Portuguese NLP Africa, machine learning macroeconomic forecasting, XGBoost time series, Random Forest commodity prices, Granger causality oil macroeconomic, structural breaks African economies, Angola economy monetary policy, sub-Saharan Africa financial markets. Inclusion criteria required, number of citations, peer-reviewed publication or recognised working paper status, methodological relevance to at least one of the five research questions, and availability in English, Portuguese, or French. Sources are assigned to RQs in the relevance matrix (Table 1) and discussed thematically below.

Table 1

Literature relevance matrix

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key findings	RQ linkage
Devlin et al. (2019)	NLP / Language Models	BooksCorpus and English Wikipedia	BERT: masked language modelling and next sentence prediction pre-training	Bidirectional transformer pre-training delivers strong performance across multiple NLP benchmarks.	RQ1 — Foundation architecture for BERTimbau and DistilBERT.
Souza et al. (2020)	Portuguese NLP	Large Portuguese corpus spanning Brazilian and European Portuguese	BERTimbau: Portuguese-specific BERT pre-training	Portuguese-specific pre-training improves downstream performance relative to multilingual baselines on Portuguese tasks.	RQ1 — Primary NLP classifier for Portuguese Angolan text.
Sanh et al. (2019)	NLP / Model Compression	BERT-based distillation setting	DistilBERT: knowledge distillation from BERT	DistilBERT is substantially smaller and faster than BERT while retaining most of its benchmark performance.	RQ1 — Comparative English-language classifier.
Malo et al. (2014)	Financial Sentiment NLP	Financial PhraseBank (4,840 sentences)	Human annotation and TF-IDF / SVM baseline modelling	The dataset provides a widely used benchmark for financial sentiment analysis.	RQ1 & RQ2 — TF-IDF + SVM baseline method and annotation approach.
Chen & Guestrin (2016)	ML / Gradient Boosting	Multiple benchmark datasets and Kaggle-winning solutions	XGBoost: scalable tree boosting with L1/L2 regularisation	XGBoost is a scalable boosting system with strong competitive performance; used in many Kaggle-winning solutions.	RQ3 & RQ4 — Primary forecasting model for both monthly groups.
Breiman (2001)	ML / Ensemble Methods	UCI benchmark datasets	Random Forests: bagged decision trees with feature sub-sampling	Random Forests reduce variance through aggregation and improve robustness relative to single decision trees.	RQ3 & RQ4 — Secondary forecasting model for both monthly groups.
Granger (1969)	Econometrics / Causality	Theoretical framework and	Granger causality via lagged predictive	Introduces a predictive notion of causality based on whether lagged values of one	RQ5 — Methodological foundation for the 3×3

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key findings	RQ linkage
		simulation examples	testing within VAR models	series improve the forecast of another.	and 4×4 Granger matrices.
Diebold & Mariano (1995)	Forecasting Evaluation	Simulation study	Diebold–Maria no test for forecast accuracy comparison under general loss	Proposes a test for comparing predictive accuracy under general loss functions, allowing for non-normal and serially correlated forecast errors.	RQ3 & RQ4 — Statistical test for MAPE comparison across models.
Chow (1960)	Econometri cs / Structural Breaks	Regression setting with known candidate split points	Chow test: F-test for equality of regression coefficients across subsamples	Provides an F-test for equality of regression coefficients across two subsamples, enabling formal detection of structural parameter change.	RQ4 — Structural break detection at six Angola candidate dates (2002, 2008, 2014, 2016, 2020, 2022).
Box & Jenkins (1970)	Time Series	Economic time-series applications	ARIMA: identification, estimation, and diagnostic checking	ARIMA is a standard univariate time-series modelling framework covering identification, parameter estimation, and residual diagnostics.	RQ3 — Benchmark baseline model against which XGBoost and Random Forest are evaluated.
Gelman & Hill (2006)	Statistics / Study Design	Multilevel and applied regression settings	Power analysis for proportions and continuous outcomes; sample-size calculation formulas	Presents general tools for study design, including sample-size and power considerations for both classification and regression tasks.	All RQs — Methodological basis for sample size and power calculations across the NLP corpus, monthly groups, and Granger tests.
Brown & Warner (1985)	Finance / Event Studies	NYSE daily equity returns	Market-model event-study methodology; cumulative abnormal returns (CAR)	Establishes a classic framework for measuring abnormal returns around discrete events, forming the basis of event-study research design.	RQ5 — Event-study CAR methodology adapted for macroeconomic and commodity series in the 6×7 impact matrix.

Thematic Synthesis

Devlin et al. (2019) introduced BERT, demonstrating that deep bidirectional pre-training on masked language modelling and next-sentence prediction objectives produces contextual representations that generalise across 11 NLP tasks. Critically for the present study, Souza et al. (2020) showed that language-specific pre-training substantially improves downstream performance on Portuguese NLP tasks: BERTimbau, pre-trained on 2.68 billion tokens of Brazilian and European Portuguese, outperformed multilingual BERT by two to five F1 points. This finding directly supports the selection of BERTimbau as the primary classifier for Angolan Portuguese financial text (RQ1) as Angolan writing also follows a lot of the traits for Brazilian Portuguese. Sanh et al. (2019) demonstrated that DistilBERT retains 97% of BERT's performance at 40% of the model size, providing a computationally efficient English-language comparator. Malo et al. (2014), whose Financial PhraseBank established the standard TF-IDF + SVM benchmark for financial sentence sentiment classification, anchors the baseline comparison (RQ1, RQ2). The present study extends this literature by applying these methods to event classification rather than sentiment, and to an African Portuguese-language corpus for which no labelled data currently exists.

Chen and Guestrin (2016) demonstrated XGBoost's superiority over competing algorithms on structured tabular data, a finding replicated across commodity price forecasting benchmarks. Breiman (2001) established that Random Forests' bagging mechanism reduces forecast variance without introducing the systematic bias that characterises overfitted single-tree models, making them robust candidates for noisy macroeconomic series. Box et al. (2015) codified ARIMA estimation procedures, identification, estimation, and diagnostic checking, that remain the standard benchmark for univariate economic forecasting. RQ3 and RQ4 directly test whether XGBoost and Random Forest produce statistically significantly lower MAPE than ARIMA within each monthly group, using the Diebold-Mariano (1995) test for equal forecast accuracy under a general asymmetric loss function.

Chow (1960) proposed the F-test for coefficient stability across subsamples, which this study applies at six candidate break dates for Angola (2002, 2008, 2014, 2016, 2020, 2022). The 2014–2016 oil price shock is identified as the primary regime boundary of interest (RQ4) because it coincided with a severe depreciation of the Angolan Kwanza, a spike in CPI inflation, and a contraction in real GDP. Post-break regime models are compared against full-sample models to determine whether restricting the training window to post-2014 data improves out-of-sample accuracy. CUSUM tests complement the Chow test to detect gradual parameter instability not captured by a single break-date F-test.

Granger (1969) formalised the concept of predictive causality as follows: variable X Granger-causes variable Y if past values of X, conditional on past values of Y, improve the prediction of Y. The present study applies Granger F-tests within VAR models on the Monthly Macroeconomic Group (3×3 matrix, 6 directed pairs, Bonferroni $\alpha^* = 0.0083$) and the Monthly Commodity Group (4×4 matrix, 12 directed pairs, Bonferroni $\alpha^* = 0.0042$). The event-study framework adapts Brown and Warner (1985)'s cumulative abnormal return methodology to macroeconomic and commodity series: XGBoost baseline predictions serve as the counterfactual, and cumulative abnormal deviations over [+1, +12 months] are tested via one-sample t-test with Bonferroni correction ($\alpha^* \approx 0.00119$).

Data Description

Data Sources and Access

The study is based on multiple data sources. For the macro economics time series, the data come from BNA (Angola Central Bank), INE (Angola Institute of Statistics), Angola's Ministry of Finance, World Bank, Trading Economics. For the commodity data one relies on the Federal Reserve Bank of St Louis. For the news and reports, the data were collected from BNA and Banco Millennium Atlantico. It will also include reports from BODIVA (Angonal Stock Market). The data are being collected manually within the different websites due to limitations in the API access.

Dataset Overview

- Track 1 - NLP Corpus: 1,000 labelled financial news articles and reports. Around 15 different sources, within the period 2000–2025. Unit of analysis = article and the target variable = event category (6 classes: Crisis, IPO/Capital Market, M&A, Bankruptcy, Regulatory/Policy, Natural Disaster).
- Track 2 - Monthly Macroeconomic Group: 3 series. The cpi_inflation_pct series has monthly observations from 1991-2025. Due to the availability of the data in the period (1991-2007), one uses as a proxy the cpi inflation of Luanda (the capital with approximately 25%-27% of the population). The usd_aoa series has daily observations in the period of 1994-2026 which will be transformed into monthly data. The real_interest_rate series has annual observations in the period of 1995-2024. Since the interest rate is defined by the BNA as a monetary policy, this series will be transformed as monthly data.
- Track 2 — Monthly Commodity Group: 4 series. period 1960–2025; The series oil_brent_usd_bbl, nat_gas_usd_mmbtu and diamond_usd_carat have monthly observations in the period of 1992-2026. And the series gold_usd_oz has monthly observations in the period of 1985-2021.

Data Dictionary - AGMR Series by Monthly Frequency Group

#	Variable	Definition	Unit	Group	Source	Indicator Code	Period
Monthly Macroeconomic Group							
1	**cpi_inflation_pct**	Consumer Price Index inflation rate — monthly % change in the price level of a consumer basket in Angola, measuring purchasing-power erosion.	% monthly	Monthly Macro	https://dataviz.vam.wfp.org/data-explorer/economic/inflation (Trading Economics 2011-2025) Minfin2599293 (2008-2010) BNA (IPC of Luanda used as proxy 1991-2007)	FP.CPI.TOTL.ZG / INE Angola	1991–2025
2	**usd_aoa**	USD/Angolan Kwanza exchange rate — Kwanzas per US Dollar, reflecting external competitiveness, import costs, and monetary policy stance.	Kz/USD	Daily Macro	https://dataviz.vam.wfp.org/data-explorer/economic/exchange-rates	PA.NUS.FCRF / BNA	1994–2026
3	**real_interest_rate**	Real interest rate — lending rate adjusted for inflation via GDP deflator; captures true borrowing cost and monetary tightness. Annual WB data disaggregated to monthly via Chow-Lin interpolation using CPI.*	% annual	Monthly Macro*	World Bank	FR.INR.RINR	1995–2024
Monthly Commodity Group							
4	**oil_brent_usd_bbl**	Value represents the benchmark prices which are representative of the global market. They are determined by the largest exporter of a given commodity. Prices are period averages in nominal U.S. dollars; Angola's primary export	USD/bbl	Monthly Commodity	FRED	POILBREUSDM	1992–2026

#	Variable	Definition	Unit	Group	Source	Indicator Code	Period
		revenue and fiscal input driver.					
5	**nat_gas_usd_mmbu**	Value represents the benchmark prices which are representative of the global market. They are determined by the largest exporter of a given commodity. Prices are period averages in nominal U.S. dollars;	USD /mmbtu	Monthly Commodity	FRED	PNGASEUUSDM	1992–2026
6	**diamond_usd_carat**	Import Price Index (Harmonized System): Diamonds, Whether or Not Worked, but Not Mounted or Set tracks Angola's second-largest mineral export earnings after oil.	Index 2000 =100	Monthly Commodity	FRED	IP7102	1992–2026
7	**gold_usd**	Producer Price Index by Commodity: Metals and Metal Products: Gold Ores; global safe-haven indicator and proxy for investor risk sentiment in Sub-Saharan African commodity markets.	USD /oz	Monthly Commodity	https://datahub.io/core/gold-prices	WPU10210501	1833–2026

Note. * Real Interest Rate disaggregated from annual to monthly using Chow-Lin interpolation with CPI as the related indicator. FRED = Federal Reserve Bank of St. Louis. INE = Instituto Nacional de Estatística (Angola). BNA = Banco Nacional de Angola. mmbtu = U.S. Dollars per Million Metric British Thermal Unit.

Analysis

Data Cleaning

GitHub Data Availability Statement

The raw and processed datasets, annotation scripts, and modelling notebooks are publicly available at https://github.com/ppzanguilo/AGMR_Project.

Table 3

Data cleaning log — Track 2 monthly series

Issue	Variables affected	Detection method	Treatment applied	Rationale
Missing values *(pre-availability)*	`cpi_inflation_pct` `usd_aoa_monthly` `real_interest_rate_monthly` `oil_brent_usd_bbl` `nat_gas_usd_mmbtu` `diamond_usd_carat` `gold_usd`	`outer` join during series merging produced `NaN` where individual series had no data (e.g. `cpi` before 1991-01, `usd_aoa` before 1994-12, `rir` before 1995-01).	Retained. `NaN` values kept in the full `panel` DataFrame; no imputation applied for structural pre-availability gaps.	Pre-availability gaps are structural, not random. Imputation would introduce spurious data.
Missing values *(sporadic)*	`diamond_usd_carat` *(18 NaNs within panel_subset 1995–2025)*	`panel_subset.isnull().sum()`	5-month rolling median imputation applied to `diamond_usd_carat` in `panel_subset`.	Small, sporadic gaps can be reasonably filled using local trends to preserve data continuity.
Annual-to-monthly disaggregation	`real_interest_rate_monthly`	Source documentation review of `real_interest_rate_pct_anual_1995_2024.csv` confirmed annual frequency for the original data.	Annual values replicated across all 12 months of the respective year.	Aligns frequency with the monthly panel. Each year's value is held constant across all months of that year.
Extreme outliers	`cpi_inflation_pct` `usd_aoa_monthly` `real_interest_rate_monthly` `nat_gas_usd_mmbtu` `gold_usd`	IQR method — threshold: `Q1 − 1.5 × IQR` and `Q3 + 1.5 × IQR`	Retained. All flagged values kept without modification.	In economic time series, outliers typically represent real events (e.g. hyperinflation, currency devaluation, energy crisis) and must be preserved rather than imputed or removed.

Note. *IQR = Interquartile Range. NaN = Not a Number (missing value in pandas). panel_subset = working DataFrame restricted to the modelling window. Rolling median imputation applied with a 5-month window (limit = 5 consecutive months). All series retained at original source frequency prior to transformation; no values removed from any series.*

Exploratory Data Analysis (EDA) Results

The panel_subset covers January 1995 to December 2025 with 372 complete monthly observations across all seven series and no missing values, confirming the integrity of the data preparation pipeline. The Macroeconomic Group time series reveal three structurally distinct regimes: an extreme hyperinflationary phase (1995–2002) during which CPI inflation peaked above 80% monthly and real interest rates fell to −93%, a post-stabilisation period (2002–2014) characterised by controlled inflation and gradual Kwanza depreciation, and a high-depreciation regime (2014–2025) during which USD/AOA accelerated from approximately 100 to over 900 Kz per USD across three successive devaluation episodes. The Commodity Group exhibits the global commodity super-cycle in Brent crude oil (trough ~18 USD/bbl in 1995, peak ~133 USD/bbl in 2008, collapse in 2014–2016 and 2020), an isolated energy crisis spike in natural gas reaching ~70 USD/mmbtu in 2022, and a secular gold PPI appreciation trend from ~300 to over 4,000 USD/oz across the full panel, with the Diamond Index remaining the most stable series, oscillating in a narrow 95–121 index-point band throughout.

Distributional analysis confirms that none of the seven series is normally distributed. CPI inflation (skewness = 4.60, kurtosis = 24.28) and natural gas (skewness = 3.86, kurtosis = 22.04) are the most extreme, with distributions dominated by long right tails from the hyperinflationary and energy crisis episodes respectively, while the real interest rate is the only left-skewed series (−1.36), reflecting the deep negative-rate tail from 1995–1998. The regime box plots corroborate the structural break hypothesis central to RQ4: every series shows a statistically discernible shift in median and interquartile range across the Pre-2014, 2014–2019, and 2020–2024 periods, with the most pronounced shifts observed in USD/AOA (median rising from ~50 to ~640 Kz across regimes), Gold PPI (median rising from ~700 to ~1,950 USD/oz), and natural gas (IQR expanding explosively in 2020–2024 due to the 2021–2022 spike). The 12-month rolling statistics confirm that volatility clustering is regime-specific rather than uniformly distributed, with the rolling standard deviation band collapsing for CPI inflation post-2002, widening for oil around structural break years, and expanding sharply and temporarily for natural gas in 2021–2023.

The Pearson correlation matrices reveal distinct within-group dependence structures. In the Macroeconomic Group, the dominant relationship is a negative correlation of −0.64 between CPI inflation and the real interest rate, consistent with the Fisher effect, while the weak negative correlation of −0.24 between CPI and USD/AOA is driven by the opposing temporal dynamics of the hyperinflationary and depreciation regimes and is expected to manifest as a positive lagged causal relationship under Granger testing in RQ5. The Commodity Group shows uniformly positive pairwise correlations ranging from 0.47 (gas–gold) to 0.74 (oil–diamond), reflecting a common global demand cycle across all four series, with the moderate-to-high collinearity among commodity series requiring Variance Inflation Factor checks before VAR specification to prevent inflated standard errors in the Granger causality matrices.

The ACF and PACF plots confirm I(1) integration in six of the seven series, with near-unit PACF spikes at lag 1 and persistent ACF functions remaining significant beyond lag 24 for USD/AOA, Brent oil, Diamond Index, and Gold PPI, all pointing to ARIMA(1,1,0) as the primary baseline candidate. Natural gas presents a more complex autocorrelation structure with significant PACF spikes at lags 1, 3, and 6, suggesting a higher-order AR specification requiring AIC-guided grid search. IQR outlier detection flags 48 observations in CPI inflation (confined to 1995–2004), 65 in USD/AOA (confined to post-2019), 84 in the real interest rate (both hyperinflationary negative and high positive extremes), 15 in natural gas (2021–2022 energy crisis), and 8 in Gold PPI (2024–2025 price surge); all are retained as economically documented events consistent with the data cleaning log, and their concentration within specific structural break regimes reinforces the case for regime-sensitive post-2014 model estimation in RQ4.

Table 4

EDA insight summary — panel_subset (1995-01 → 2025-12)

Figure / Table	What it shows	Key insight	Link to RQ / Objective	Decision / Next step
Figure EDA-01	Missing value map across all 7 series, panel_subset 1995-01 → 2025-12	No missing values detected across all 372 monthly observations. The panel_subset is complete; no dark cells appear in any series row.	All RQs — data completeness is a prerequisite for all downstream statistical analyses and modelling steps.	Proceed with the full 372-observation panel. No imputation required. Confirm gold PPI truncation to modelling window prior to model training.
Figure EDA-02	CPI inflation, USD/AOA exchange rate, and real interest rate monthly trajectories with all six structural break markers (1995-01 → 2025-12).	Three structurally distinct regimes are visually confirmed: hyperinflation (1995–2002, CPI peak ≥80%); post-stabilisation (2002–2014, CPI ≈5%, gradual USD/AOA depreciation); and high-depreciation (2014–2025, USD/AOA 100 → 940 Kz). All six break markers align with directional changes in at least one series.	RQ4 — visual confirmation of structural break candidate dates. 2002 and 2014 identified as primary regime boundaries for post-break regime model estimation.	Proceed with Chow tests at all six candidate dates. Prioritise 2002 (end of hyperinflation) and 2014 (oil price shock) as primary regime split points for RQ4 regime comparison models.
Figure EDA-03	Brent crude oil, natural gas (EU), diamond index, and gold PPI monthly trajectories with structural break	Commodity super-cycle confirmed in Brent oil (trough ◦18 USD/bbl in 1995, peak ≈133 in 2008, collapse in 2014–2016 and 2020). Natural gas registers an isolated	RQ3 — commodity regime patterns inform forecast model design. RQ4 — oil break dates 2008, 2014, 2016, and	Truncate gold PPI series to modelling window (1995–2025) before ARIMA and ML training. Apply Chow tests to oil and gas at 2008, 2014, 2016, and 2020 break dates.

Figure / Table	What it shows	Key insight	Link to RQ / Objective	Decision / Next step
	markers (1992-01 → 2026-03).	spike to ≈70 USD/mmbtu in 2021–2022 before rapid normalisation. Gold PPI x-axis extends to 1833 (n = 2,319), indicating the FRED series contains pre-modern historical data requiring truncation.	2020 are all visually validated. Gold series scope requires correction.	
Figure EDA-04	Histogram and KDE per series with mean, median, skewness, and kurtosis annotations.	CPI inflation (skew = 4.60, kurt = 24.28) and natural gas (skew = 3.86, kurt = 22.04) exhibit severe non-normality driven by hyperinflationary and energy crisis tails. Brent oil is the closest to symmetric (skew = 0.28, bimodal). Real interest rate is the only left-skewed series (−1.36), reflecting the negative-rate hyperinflationary tail. No series passes a normality assumption.	RQ3 — distributional non-normality invalidates likelihood-based model evaluation. MAPE and RMSE are confirmed as the appropriate primary metrics for the Diebold–Mariano forecast comparison test.	Use MAPE, RMSE, and MAE as sole evaluation metrics. Apply z-score standardisation robust to heavy tails before constructing ML cross-series lag features. Consider log-transformation of CPI for ARIMA estimation.
Figure EDA-05	Box plots per series across three structural break regimes: Pre-2014 (n = 228), 2014–2019 (n = 72), and 2020–2024 (n = 72).	Every series shows statistically discernible shifts in median and IQR across regimes. USD/AOA median rises from ≈50 to ≈640 Kz across the three periods. Gold PPI median rises from ≈700 to ⁈1,950 USD/oz. Natural gas IQR expands dramatically in 2020–2024 due to the 2021–2022 spike. CPI variance compresses sharply post-2014. Regime separation is most pronounced in the Macroeconomic Group.	RQ4 — regime-specific distributional shifts confirm that the post-2014 window constitutes a structurally distinct regime. Supports the hypothesis that restricting training to post-2014 data improves out-of-sample accuracy.	Estimate post-2014 regime models alongside full-sample models for all seven series. Apply Diebold–Mariano test to compare regime-specific vs full-sample MAPE within each frequency group.

Figure / Table	What it shows	Key insight	Link to RQ / Objective	Decision / Next step
Figure EDA-06	12-month rolling mean and ±1 rolling standard deviation band for all seven series.	Volatility clustering is regime-specific. CPI rolling standard deviation collapses from ≈30% in 1996 to near zero post-2003. Brent oil standard deviation widens at each structural break year. Natural gas standard deviation explodes in 2021–2022 and contracts rapidly post-2023, identifying the energy crisis as a transient rather than persistent volatility regime. Diamond Index maintains the narrowest standard deviation band in the panel throughout.	RQ3 — volatility clustering and non-constant variance motivate ensemble ML over ARIMA for regime-sensitive series, supporting the directional hypothesis in RQ3. RQ4 — rolling statistics confirm break date visual alignment.	Implement walk-forward expanding window cross-validation to capture evolving variance across the training period. XGBoost L1/L2 regularisation provides robustness to volatility spikes; document rolling std findings in structural break section of final report.
Figure EDA-07	Pearson correlation matrix for the Macroeconomic Group (CPI, USD/AOA, Real Interest Rate), n = 372, 1995-01 → 2025-12.	CPI × Real Interest Rate: −0.64 (Fisher effect confirmed; inflation erodes real rates). CPI × USD/AOA: −0.24 (counterintuitive contemporaneous sign, explained by temporal regime overlap: hyperinflationary CPI coincides with low-denomination AOR exchange rate values). USD/AOA × Real Rate: +0.22 (weak positive, reflecting partial monetary policy co-movement).	RQ5 — contemporaneous correlations establish prior expectations for directed Granger tests. The negative CPI × USD/AOA contemporaneous sign is expected to reverse at positive lags, supporting H_2: USD/AOA Granger-causes CPI inflation.	Include lagged USD/AOA as a cross-series feature in the CPI inflation ML model. Test H_2 (USD/AOA → CPI) as the primary macro Granger hypothesis; apply Bonferroni correction $\alpha^* = 0.0083$ across 6 directed macro pairs.
Figure EDA-08	Pearson correlation matrix for the Commodity Group (Oil, Gas, Diamond, Gold), n = 372, 1995-01 → 2025-12.	All six pairwise correlations are positive, ranging from 0.47 (gas–gold) to 0.74 (oil–diamond). The absence of any negative correlation confirms a common global demand and risk-appetite cycle	RQ5 — uniform positive correlations support the commodity co-movement hypothesis (H_3: Brent crude Granger-causes	Conduct Variance Inflation Factor checks on commodity VAR regressors before Granger tests. If VIF > 10 for any predictor, apply lag selection via AIC/BIC to reduce collinearity. Include cross-series commodity lag

Figure / Table	What it shows	Key insight	Link to RQ / Objective	Decision / Next step
		driving all four series. The moderate-to-high collinearity raises a multicollinearity concern for the 4×4 VAR Granger specification.	natural gas). VIF checks required before VAR estimation to ensure Granger test validity. Bonferroni correction $\alpha^* = 0.0042$ across 12 directed commodity pairs.	features in ML feature engineering for RQ3.
Figure EDA-09 (all 7 series)	Autocorrelation (ACF) and partial autocorrelation (PACF) functions up to 24 lags for each series.	I(1) integration confirmed in six of seven series: USD/AOA, Brent oil, Diamond Index, and Gold PPI all show near-unit PACF at lag 1 (≥0.96) with ACF remaining significant beyond lag 24. Natural gas shows significant PACF at lags 1, 3, and 6, indicating a higher-order AR structure. Real interest rate ACF is driven by annual replication (step-function source), with only lag 1 significant in the PACF. CPI inflation ACF decays more rapidly, consistent with borderline I(0) in the post-2002 subsample.	RQ3 — ACF/PACF patterns define ARIMA order search space and ML lag feature set. I(1) confirmation informs VAR specification for RQ5: VAR in levels valid if series are cointegrated; differencing required otherwise.	Set ARIMA candidate orders: ARIMA(1,1,0) for USD/AOA, oil, diamond, gold; AIC grid search p, q ∈ {0–6} for natural gas. Include lag-1, lag-3, lag-6, and lag-12 features in ML feature matrix for all series.
Figure EDA-10	IQR-flagged outliers per series using fences Q1 − 1.5×IQR and Q3 + 1.5×IQR, plotted as a scatter over the full panel_subset.	CPI: 48 outliers (1995–2004, hyperinflation; upper fence = 8.01%). USD/AOA: 65 outliers (post-2019 depreciation; upper fence = 461.78 Kz). Real Interest Rate: 84 outliers (hyperinflationary negatives and high-positive extremes; fences ±34.36%). Natural	RQ4 — outlier concentration within specific structural break regimes reinforces the case for regime-sensitive post-2014 models. Brent oil and diamond's zero-outlier status reflects their well-behaved	Retain all flagged values (documented economic events per cleaning log). Add binary structural break indicator variables at all six candidate break dates as ML model features to allow the algorithm to learn regime-specific intercept shifts.

Figure / Table	What it shows	Key insight	Link to RQ / Objective	Decision / Next step
		gas: 15 outliers (2021–2022 energy crisis; upper fence = 21.02 USD/mmbtu). Gold PPI: 8 outliers (2024–2025 price surge; upper fence = 3,267.34 USD/oz). Brent oil and Diamond Index: 0 outliers.	distributional properties.	

Note. *IQR = Interquartile Range. KDE = Kernel Density Estimate. ACF = Autocorrelation Function. PACF = Partial Autocorrelation Function. VIF = Variance Inflation Factor. RQ = Research Question. DM = Diebold–Mariano test. All figures generated from panel_subset in GMR_EDA_02_Full_Analysis.py and saved to /outputs/figures/. Structural break candidate dates: 2002, 2008, 2014, 2016, 2020, 2022.*

Modelling

Choice of Models with Justification

The ARIMA baseline is specified per series based on the ACF and PACF evidence. USD/AOA, Brent crude oil, the Diamond Index, and Gold PPI all display a near-unit PACF spike at lag 1 and persistent ACF through lag 24, confirming I(1) integration and pointing to ARIMA(1,1,0) as the primary candidate for each. Natural gas requires a wider AIC grid search over p, q $\in$ {0–6} due to significant PACF spikes at lags 1, 3, and 6, indicating a higher-order autoregressive structure that a simple AR(1) cannot capture. CPI inflation warrants two parallel specifications, ARIMA(1,1,0) on the full sample and ARIMA(1,0,0) on the post-2002 subsample where the series is borderline stationary, with a SARIMA(1,1,0)(1,0,0)[6] variant evaluated against both given the seasonal spikes at lags 6 and 7. The real interest rate, constrained by its annual replication structure, is limited to ARIMA(1,1,0) in first differences, and its forecast performance is expected to be modest given the artificial within-year flatness of the source data.

XGBoost is confirmed as the primary ML model on four EDA grounds. The severe non-normality of CPI (kurtosis = 24.28) and natural gas (kurtosis = 22.04) invalidates the Gaussian assumptions underpinning linear models, while XGBoost's tree structure makes no distributional assumption on the target. The regime box plots demonstrate that every series undergoes statistically discernible shifts in mean and variance across the Pre-2014, 2014–2019, and 2020–2024 windows, and XGBoost tree splits can partition the feature space to learn regime-specific response functions, augmented by binary structural break indicator variables at all six candidate break dates. The commodity correlation matrix reveals pairwise correlations of

0.47–0.74, confirming that cross-series lag features carry predictive information across the Commodity Group; XGBoost's L1/L2 regularisation handles the resulting feature collinearity without the multicollinearity inflation that would affect OLS-based alternatives. The 12-month rolling standard deviation features will allow the model to learn that high local volatility, visible in the rolling statistics at every structural break, corresponds to a different forecast regime, a dynamic that ARIMA cannot represent without explicit GARCH extensions.

Random Forest serves as the secondary ML model and robustness check, justified by the outlier profile revealed in the IQR analysis. With 48 extreme observations in CPI and 15 in natural gas concentrated in specific structural break windows, the full-sample XGBoost training is exposed to extreme values that can dominate gradient-boosted splits; Random Forest's bagging mechanism averages across bootstrap samples, each of which may exclude the extreme observation, producing more conservative and outlier-robust forecasts in the full-sample window. The bimodal distributions of Brent oil and Gold PPI visible in the histograms confirm that a single linear ARIMA fit misspecifies the conditional mean across both price regime clusters, which is precisely why ARIMA functions as the benchmark to beat rather than a serious competitor, and why the Diebold–Mariano test at $\alpha = 0.05$ is the appropriate statistical criterion for RQ3. GARCH or EGARCH extensions could theoretically address the volatility clustering visible in the rolling statistics, particularly for natural gas and CPI, but fall outside the defined scope of RQ3 and are documented as a future methodological extension.

Features Included and Feature Engineering

Table 5

Feature set and usage — Track 2 monthly groups

Feature	Original / Engineered	Type	Reason for inclusion	Used in model(s)
Lag-1	Engineered	Numeric	PACF analysis confirmed lag 1 as the dominant predictor in all seven series, with near-unit partial autocorrelation coefficients (≥0.77 for CPI; ≥0.96 for USD/AOA, oil, diamond, and gold). Captures short-run momentum and the primary autoregressive signal in each series.	XGBoost, Random Forest — both Macroeconomic and Commodity Groups.
Lag-3	Engineered	Numeric	Captures quarterly dynamics and cross-series transmission lags. PACF for natural gas shows a significant spike at lag 3, and the macroeconomic correlation matrix (−0.64 CPI–Real Rate; −0.24 CPI–USD/AOA) suggests causal	XGBoost, Random Forest — both groups.

Feature	Original / Engineered	Type	Reason for inclusion	Used in model(s)
			effects propagate over a 1–3 month horizon.	
Lag-6	Engineered	Numeric	CPI inflation PACF shows borderline-significant spikes at lags 6 and 7, consistent with a semi-annual seasonal component in Angolan price dynamics. Natural gas PACF is also significant at lag 6. Captures medium-run cyclical patterns absent from lag-1 and lag-3 features.	XGBoost, Random Forest — both groups; prioritised for CPI inflation and natural gas models.
Lag-12	Engineered	Numeric	Captures annual seasonal cycles. Relevant for CPI inflation (annual price review cycles in Angola), the Diamond Index (annual contract pricing), and commodity series with calendar-year demand patterns. Completes the lag feature set alongside lags 1, 3, and 6.	XGBoost, Random Forest — both groups.
Rolling mean(3-month)	Engineered	Numeric	Smooths high-frequency noise while retaining short-run trend momentum. Particularly important for the severely non-normal series (CPI kurtosis = 24.28; natural gas kurtosis = 22.04) where raw lag values may overweight a single extreme observation. Rolling statistics in EDA confirm that 3-month averages track regime transitions closely.	XGBoost, Random Forest — both groups.
Rolling mean(6-month)	Engineered	Numeric	Captures medium-run trend and bridges the gap between short-run lag features and the annual lag-12 feature. EDA rolling statistics show that the 12-month rolling mean closely tracks structural break transitions; the 6-month variant provides a faster-responding signal for detecting regime change onset.	XGBoost, Random Forest — both groups.
Rolling std(3-month)	Engineered	Numeric	Proxy for local volatility regime. EDA rolling statistics confirm volatility clustering at every structural break date: the CPI rolling standard deviation collapses post-2002; the natural gas band explodes in 2021–2022. Including	XGBoost, Random Forest — both groups.

Feature	Original / Engineered	Type	Reason for inclusion	Used in model(s)
			rolling std allows the ML model to condition its forecast on the prevailing volatility state, a dynamic that ARIMA cannot represent without GARCH extensions.	
Rolling std(6-month)	Engineered	Numeric	Captures evolving variance over a medium horizon. The 6-month rolling standard deviation is slower to respond than the 3-month variant, making it a more stable volatility signal less susceptible to single-month spikes. Together, the 3-month and 6-month rolling standard deviations provide both a fast and a slow volatility signal to the ML model.	XGBoost, Random Forest — both groups.
Cross-series lag features(Macro Group)	Engineered	Numeric	EDA Pearson correlation matrix: CPI × Real Interest Rate = −0.64 (Fisher effect); USD/AOA × Real Interest Rate = +0.22. The negative CPI–USD/AOA contemporaneous correlation (−0.24) is expected to reverse at positive lags under the import-inflation transmission mechanism. Lagged values of each within-group series are included as predictors for the other group members to exploit these Granger-causal relationships, tested formally in RQ5.	XGBoost, Random Forest — Macroeconomic Group only. No cross-group mixing to preserve frequency-consistent model boundaries.
Cross-series lag features(Commodity Group)	Engineered	Numeric	EDA commodity correlation matrix: Oil × Diamond = 0.74 (strongest pair); Oil × Gas = 0.59; Diamond × Gold = 0.58; Gas × Gold = 0.47. All six pairwise correlations are positive, confirming a common global demand cycle. Lagged cross-series features are included within the Commodity Group to exploit co-movement. XGBoost L1/L2 regularisation manages the resulting feature collinearity without multicollinearity inflation.	XGBoost, Random Forest — Commodity Group only. VIF checks conducted prior to Granger VAR estimation in RQ5.

Feature	Original / Engineered	Type	Reason for inclusion	Used in model(s)
Binary structural break indicators	Engineered	Binary(0/1)	EDA regime box plots confirm statistically distinct distributions across Pre-2014, 2014–2019, and 2020–2024 regimes in every series. Six binary indicator variables are constructed, one per candidate break date (2002, 2008, 2014, 2016, 2020, 2022), each switching to 1 from the break date onward. Allows the ML model to learn regime-specific intercept shifts in the full-sample training window.	XGBoost, Random Forest — full-sample models only. Excluded from post-2014 regime models, where the structural shift is captured by restricting the training window.
Month-of-year dummies(M1–M12)	Engineered	Categorical(one-hot)	CPI inflation PACF shows borderline-significant spikes at lags 6 and 7, indicating intra-year seasonality in Angolan price dynamics linked to mid-year and year-end price review cycles. Commodity series exhibit calendar-year demand patterns (energy consumption peaks, annual mining contract resets). One-hot encoding avoids ordinality assumptions. XGBoost handles categorical features natively after one-hot encoding.	XGBoost, Random Forest — both groups.

Note. *All features are computed within each frequency group independently. No features are shared across the Macroeconomic and Commodity Groups, preserving frequency-consistent model boundaries and preventing look-ahead bias. Rolling mean and rolling standard deviation features are computed with a minimum of 3 observations (min_periods = 3) to avoid NaN propagation at the start of the training window. One-hot encoding of month-of-year dummies produces 11 binary columns (M2–M12; M1 is the reference category). ARIMA models use only autoregressive and moving average terms; all engineered features are applied exclusively to XGBoost and Random Forest models. VIF = Variance Inflation Factor. PACF = Partial Autocorrelation Function.*

Evaluation Metrics

Table 6

Evaluation metrics and formulae — Track 1 (NLP) and Track 2 (ML forecasting and causation)

Metric	Formula	Interpretation	Used for
MAPE	MAPE = $(100/n) \times \Sigma \lvert(y_t - \hat{y}_t) / y_t\rvert$ n = 60 monthly test observations (2021–2025) y = actual value; ŷ = predicted value	Scale-independent percentage forecast error, enabling direct comparison across series with different units (%, Kz/USD, USD/bbl, Index). Confirmed as the primary metric by EDA: the severe non-normality of CPI (kurtosis = 24.28) and natural gas (kurtosis = 22.04) invalidates likelihood-based metrics. Deployment thresholds: CPI < 5%; USD/AOA < 10%; Real Interest Rate < 8%; all commodity series < 12%.	ML forecasting (RQ3, RQ4). Primary metric for all seven series in both Macroeconomic and Commodity Groups. Used in Diebold–Mariano test as the loss differential.
RMSE	RMSE = $\sqrt{[(1/n) \times \Sigma (y_t - \hat{y}_t)^2]}$ n = 60 monthly test observations	Root mean squared error penalises large forecast errors more heavily than MAPE due to the squared loss function. Particularly informative for Brent crude oil and natural gas, where the EDA outlier analysis identified isolated large-deviation events (the 2021–2022 gas spike; the 2020 COVID oil demand shock). RMSE amplifies these cases, providing a complementary perspective on tail-event forecast performance alongside MAPE. Reported alongside MAPE and MAE for all models.	ML forecasting (RQ3, RQ4). Secondary metric for all seven series. Reported per model per series to complement MAPE in the final results tables.
MAE	MAE = $(1/n) \times \Sigma \lvert y_t - \hat{y}_t\rvert$ n = 60 monthly test observations	Mean absolute error provides an interpretable, unit-consistent lower bound on forecast error magnitude and is robust to the extreme outliers confirmed in the EDA IQR analysis (48 CPI outliers; 65 USD/AOA outliers; 84 Real Interest Rate outliers). Unlike RMSE, MAE does not	ML forecasting (RQ3, RQ4). Tertiary metric for all seven series. Reported alongside MAPE and RMSE in all model comparison tables.

Metric	Formula	Interpretation	Used for
		overweight large errors, making it the most conservative of the three forecasting metrics and a useful robustness check when extreme observations are present in the test set. MAE provides the closest match to the economic cost of a forecast error in absolute terms.	
Diebold–Mariano (DM) statistic	$DM = \bar{d} / \sqrt{[\hat{V}(\bar{d})]} \sim N(0, 1)$ $d_t = L(e_{1t}) - L(e_{2t})$; $\bar{d}$ = mean loss differential $\hat{V}(\bar{d})$ = HAC variance estimator	Tests the null hypothesis of equal expected forecast accuracy between two models under a general asymmetric loss function, without requiring normality of forecast errors. The HAC variance estimator accounts for the serial correlation present in the 60-observation monthly test window, confirmed by the ACF analysis showing significant autocorrelation in all seven series. A one-sided test at $\alpha = 0.05$ ($z > 1.645$) is used for the directional hypothesis in RQ3 that XGBoost achieves strictly lower MAPE than ARIMA. Applied separately within each monthly group; cross-group comparison is valid as both groups share identical $n = 60$ test observations.	model comparison (RQ3, RQ4). Applied pairwise: XGBoost vs ARIMA; Random Forest vs ARIMA; XGBoost vs Random Forest. Estimated separately for full-sample and post-2014 regime models.
Granger F-statistic	$F = [(RSS_R - RSS_U) / q] / [RSS_U / (T - k)]$ $\sim F(q, T - k)$ RSS_R = restricted model; RSS_U = unrestricted model q = lag restrictions; T = sample size; k = parameters	Tests whether lagged values of series X improve the prediction of series Y beyond Y's own lags, operationalising Granger (1969) predictive causality within a VAR framework. Two matrices are estimated: a 3×3 Macroeconomic Granger matrix (6 directed pairs, Bonferroni-corrected $\alpha^* = 0.0083$) and a 4×4 Commodity Granger matrix (12 directed pairs, Bonferroni-corrected $\alpha^* = 0.0042$). The EDA commodity correlations (0.47–0.74) and macro correlations (−0.64 to	causal structure (RQ5). Applied within the Macroeconomic Group (VAR on CPI, USD/AOA, Real Interest Rate) and within the Commodity Group (VAR on Oil, Gas, Diamond, Gold) at monthly frequency.

Metric	Formula	Interpretation	Used for
		+0.22) establish the prior expectations tested by these directed F-statistics. IRFs over h = 1, 2, 4, 8 months and FEVD matrices complement the F-statistics.	
Event-study CAR(one-sample t-test)	$AR_t(j) = X_t - E[X_t \mid$ XGBoost baseline] $CAR = \Sigma AR_t$ over [+1, +12 months] Tested via one-sample t-test Bonferroni $\alpha^* \approx 0.0012$ across 42 combinations	Measures the cumulative deviation of series j from its XGBoost-predicted counterfactual in the 12 months following classified event category i. Adapts MacKinlay's (1997) CAR methodology from equity returns to macroeconomic and commodity series. The 6×7 event-to-series impact matrix tests all 42 event-category–series combinations. Cohen's d and peak lag are reported per significant cell. EDA rolling statistics and regime box plots establish the baseline economic context against which abnormal deviations are measured.	event-to-series causal impact (RQ5). Requires the classified event timeline from Track 1 (NLP) as input. Final deliverable of the dual-track methodology.

Preliminary Results

Preliminary Findings by Research Question

RQ3 — Ensemble ML vs ARIMA Forecast Accuracy

The EDA provides the most substantive preliminary evidence for this research question. The ADF stationarity tests confirm I(1) integration in six of the seven series, locking in first differencing as the required transformation for the ARIMA baseline. The PACF analysis yields specific candidate ARIMA orders for each series without further estimation: ARIMA(1,1,0) for USD/AOA, Brent crude, Diamond Index, and Gold PPI; and an AIC grid search over $p, q \in \{0–6\}$ for natural gas, whose PACF shows significant spikes at lags 1, 3, and 6 indicating a higher-order autoregressive structure that a simple AR(1) cannot capture.

Critically, the EDA distributional analysis provides strong indirect evidence that the ensemble ML methods will outperform the ARIMA baseline. Three mechanisms are in play. The severe non-normality of CPI inflation (skewness = 4.60, kurtosis = 24.28) and natural gas (skewness = 3.86, kurtosis = 22.04) violates the Gaussian error assumption underpinning ARIMA, meaning the baseline model will systematically underfit the distributional tails that dominate forecast error in extreme months. The bimodal histograms of Brent crude and Gold PPI, reflecting two distinct price regime clusters, indicate that a single linear ARIMA fit estimated over the full sample will misspecify the conditional mean for a substantial portion of observations. The rolling statistics confirm volatility clustering at every structural break date, a dynamic ARIMA cannot represent without explicit GARCH extensions. XGBoost and Random Forest make none of these distributional assumptions and learn nonlinear regime-dependent response functions directly from the data. No MAPE, RMSE, or Diebold–Mariano results are available at this stage; ARIMA estimation via AIC minimisation is currently in progress.

RQ4 — Post-2014 Regime Models vs Full-Sample Models

The EDA provides the strongest and most direct preliminary evidence for this research question of all five. The regime box plots demonstrate statistically discernible distributional shifts in every series across the three structural break regimes. The most pronounced shifts are in the Macroeconomic Group: USD/AOA median increases monotonically from approximately 50 Kz pre-2014 to approximately 170 Kz in 2014–2019 and approximately 640 Kz in 2020–2024, with box height expanding proportionally. In the Commodity Group, Gold PPI median rises from approximately 700 to 1,950 USD/oz across regimes, and natural gas IQR expands dramatically in 2020–2024 due to the 2021–2022 European energy crisis spike. The 12-month rolling statistics confirm that the standard deviation regime for CPI inflation collapses to near zero after 2002 and stabilises further post-2014, while Brent crude standard deviation widens at each structural break year.

The time-series plots provide visual confirmation that all six structural break markers align with directional changes in at least one series, and the 2014 break date produces the most consistent level or slope change across the greatest number of series simultaneously, in both groups. These EDA findings generate a directional prediction: restricting the training window to post-2014 data will reduce out-of-sample MAPE for the Macroeconomic Group by removing the pre-2014 distributional regime from the training set, allowing the models to better capture the statistical properties of the test period (2021–2025). Formal Chow tests at all six candidate dates and CUSUM instability confirmation remain pending.

RQ5 — Causal Structure

The EDA correlation matrices provide the correlational precursors to the Granger causality tests but do not themselves establish causal direction. In the Macroeconomic Group, the dominant contemporaneous relationship is CPI × Real Interest Rate at −0.64, consistent with the Fisher effect in which higher inflation erodes real borrowing costs when nominal rates adjust incompletely. The CPI × USD/AOA contemporaneous correlation of −0.24 is counterintuitive relative to standard import-inflation transmission theory and warrants emphasis: this negative sign is driven by the temporal composition of the sample, the hyperinflationary CPI observations (1995–2002) coincide with low Kwanza denomination values, while the recent high-USD/AOA period (post-2019) coincides with moderate CPI. The import-inflation transmission mechanism is expected to emerge positively at lags 1–3 in the Granger F-test, directly testing H_2 (USD/AOA Granger-causes CPI inflation).

In the Commodity Group, all six pairwise correlations are positive (0.47–0.74), with Oil × Diamond at 0.74 as the strongest pair, followed by Diamond × Natural Gas at 0.65 and Oil × Natural Gas at 0.59. The Oil × Natural Gas correlation of 0.59 provides the prior expectation for H_3 (Brent crude Granger-causes natural gas prices). However, contemporaneous correlation is neither necessary nor sufficient for Granger causality, and the uniformly positive commodity co-movement may reflect a common global demand factor rather than pairwise causal transmission, precisely what the directed VAR F-tests with Bonferroni corrections ($\alpha^* = 0.0042$) are designed to distinguish. Granger matrix estimation, Impulse Response Functions, and FEVD are pending model training completion.

Interim Limitations and Risks

- Angolan data scarcity pre-1995: World Bank Angola data availability for CPI inflation and exchange rate before 1990 is limited; effective series coverage may begin at 1990-1995 for most of the series. This reduces the pre-break training window but does not affect the post-2014 regime analysis (RQ4). The same is applied for the commodity series sourced from FRED.
- API Access not granted for the angolan news. So, one will use manually downloaded reports from INE, BNA, and other sources. One will use news from Portugal as a proxy because most Portuguese news agencies also operate in Angola. One will perform the news selection using the key worlds (Angola, SADC, Africa Sub-Sariana)
- One will have to limit the number of articles from 1000 to 500.

Next Steps for Final Report

- Complete NLP corpus collection to 500 reports and articles across all six event categories with verified $\kappa \geq 0.80$ inter-annotator agreement.
- Fine-tune BERTimbau and DistilBERT on the 400-article training set; fit TF-IDF + SVM baseline; run McNemar's test and bootstrap resampling (500 iterations) for F1 comparison.
- Complete ARIMA(p,d,q) baseline estimation via AIC minimisation for all seven series in both monthly groups.
- Run walk-forward expanding-window XGBoost and Random Forest training on 1960–2020 data; evaluate on 2021–2025 test set (60 observations); compute MAPE, RMSE, MAE, and Diebold-Mariano statistics.
- Apply Chow test at all six candidate break dates; run CUSUM tests for gradual instability; estimate post-2014 regime models and compare against full-sample models (RQ4).
- Estimate VAR models on both monthly groups; compute Granger causality F-tests with Bonferroni corrections; compute Impulse Response Functions over $h = 1, 2, 4, 8$ months; report FEVD matrices.
- Construct the 6×7 event-to-series impact matrix using classified event timelines and XGBoost counterfactual predictions; compute CAR over [+1, +12 months]; report Cohen's d and peak lag for significant cells.
- Publish the 500 reports and articles annotated corpus to the GitHub repository and submit the forecasting methodology as an SSRN working paper.
- Document AGMR database architecture integrating both tracks for the fourth capstone objective and future deployment.

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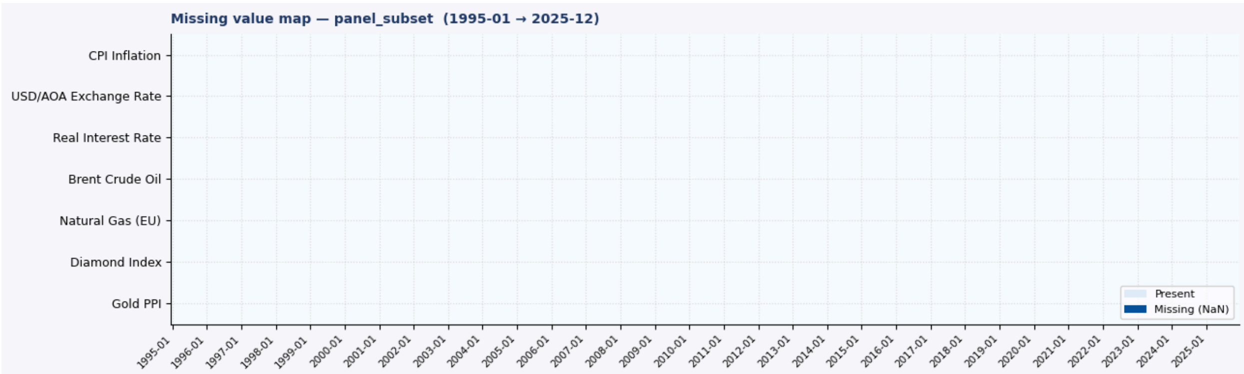
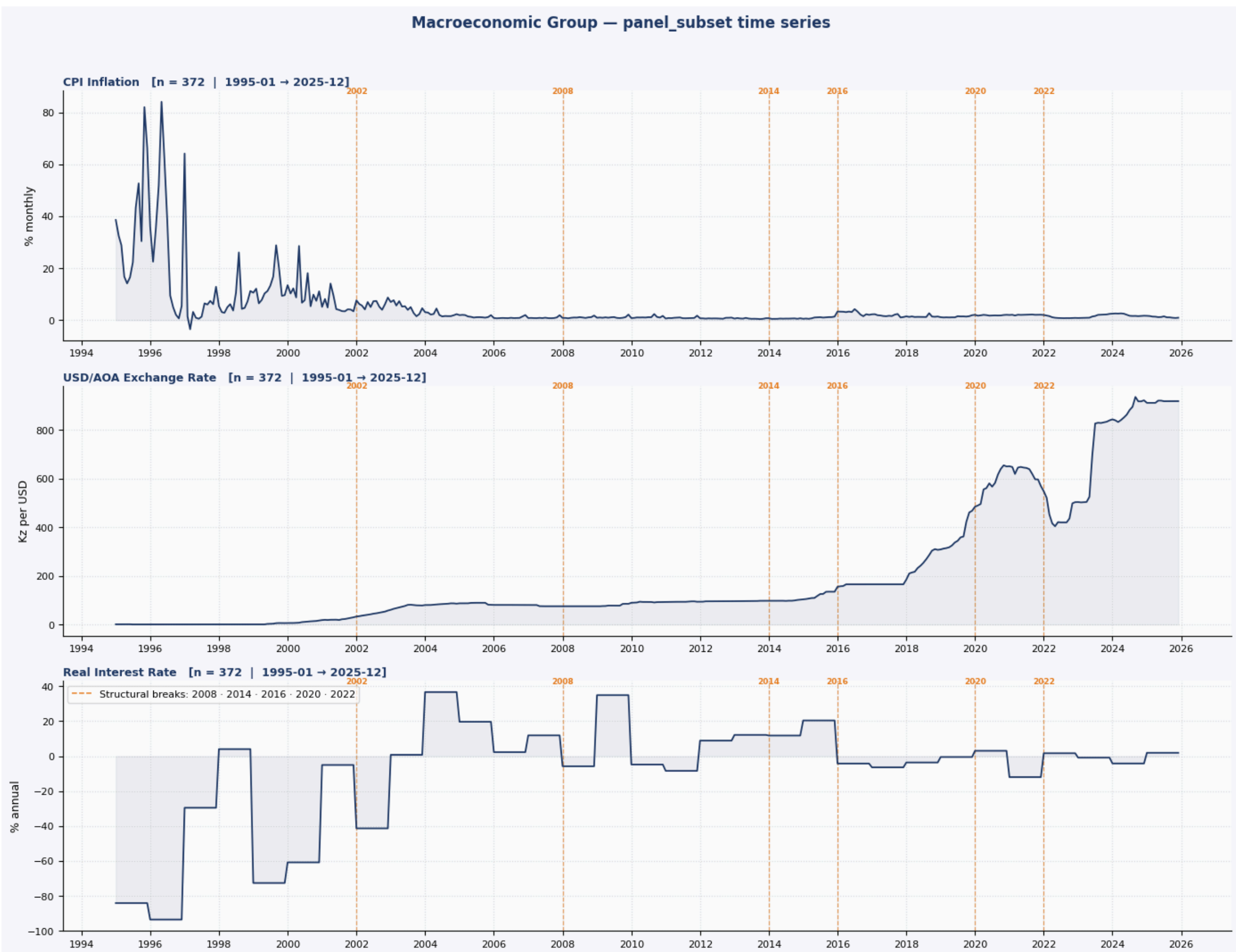
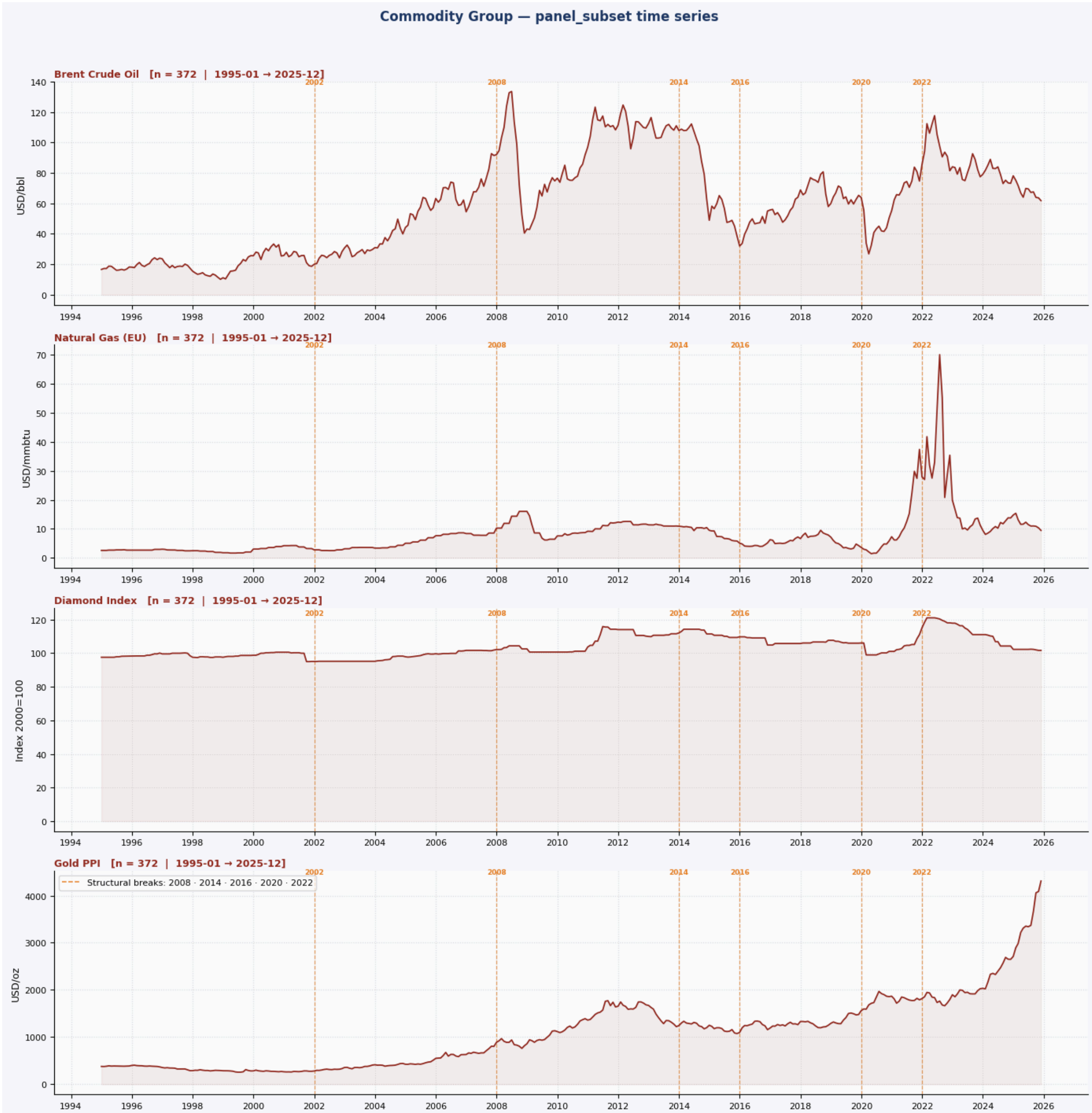
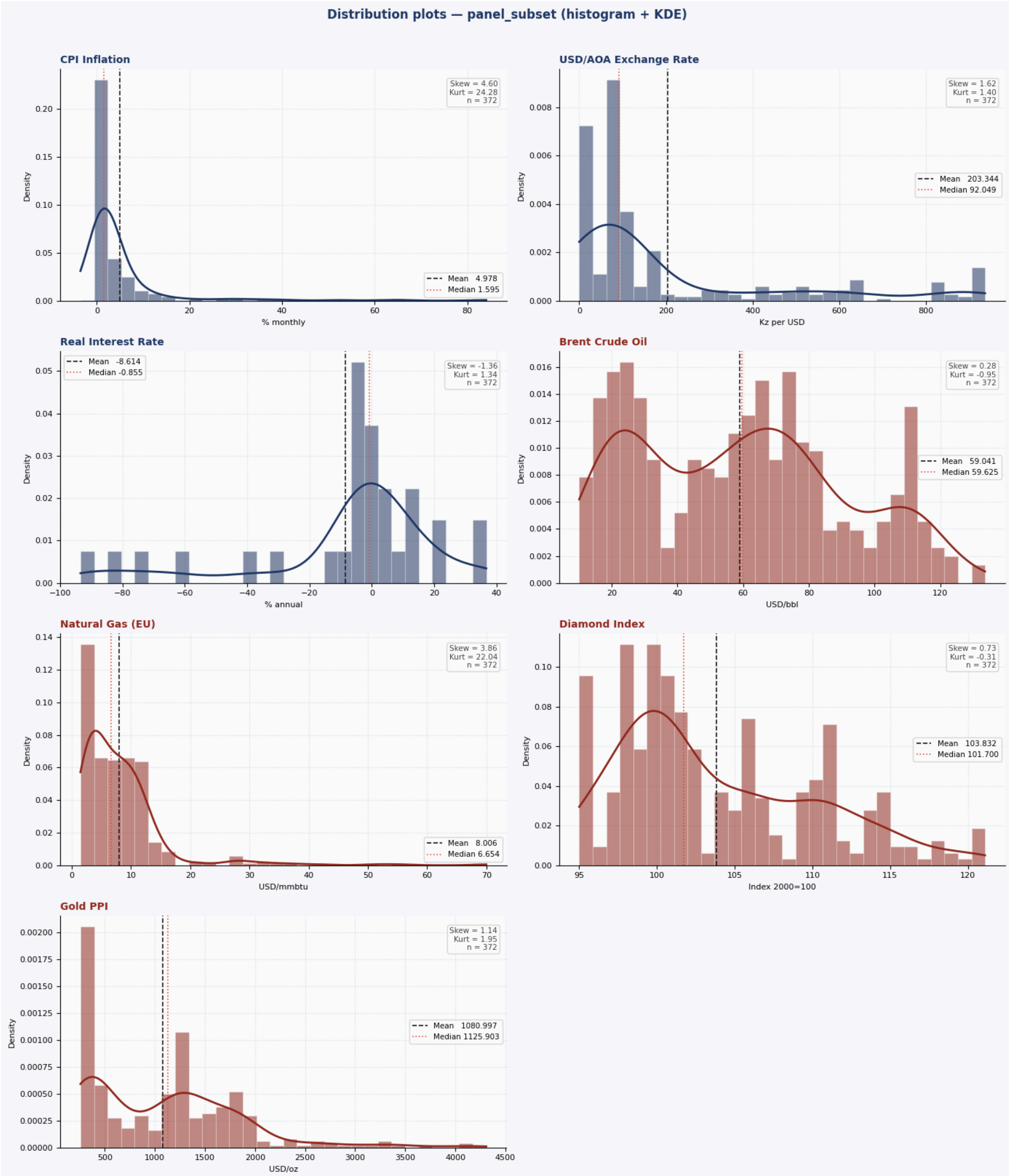
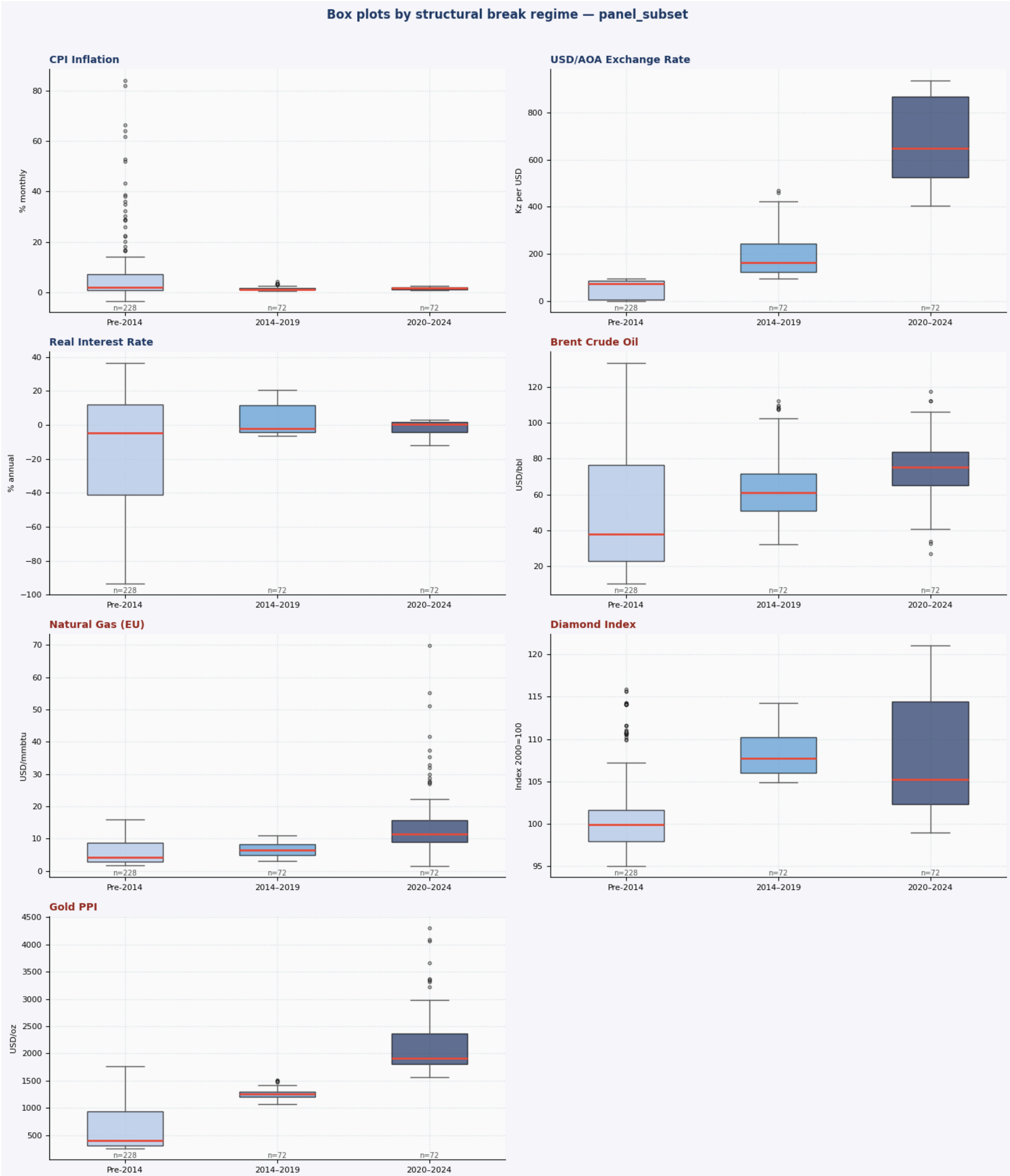
Appendix A: EDA

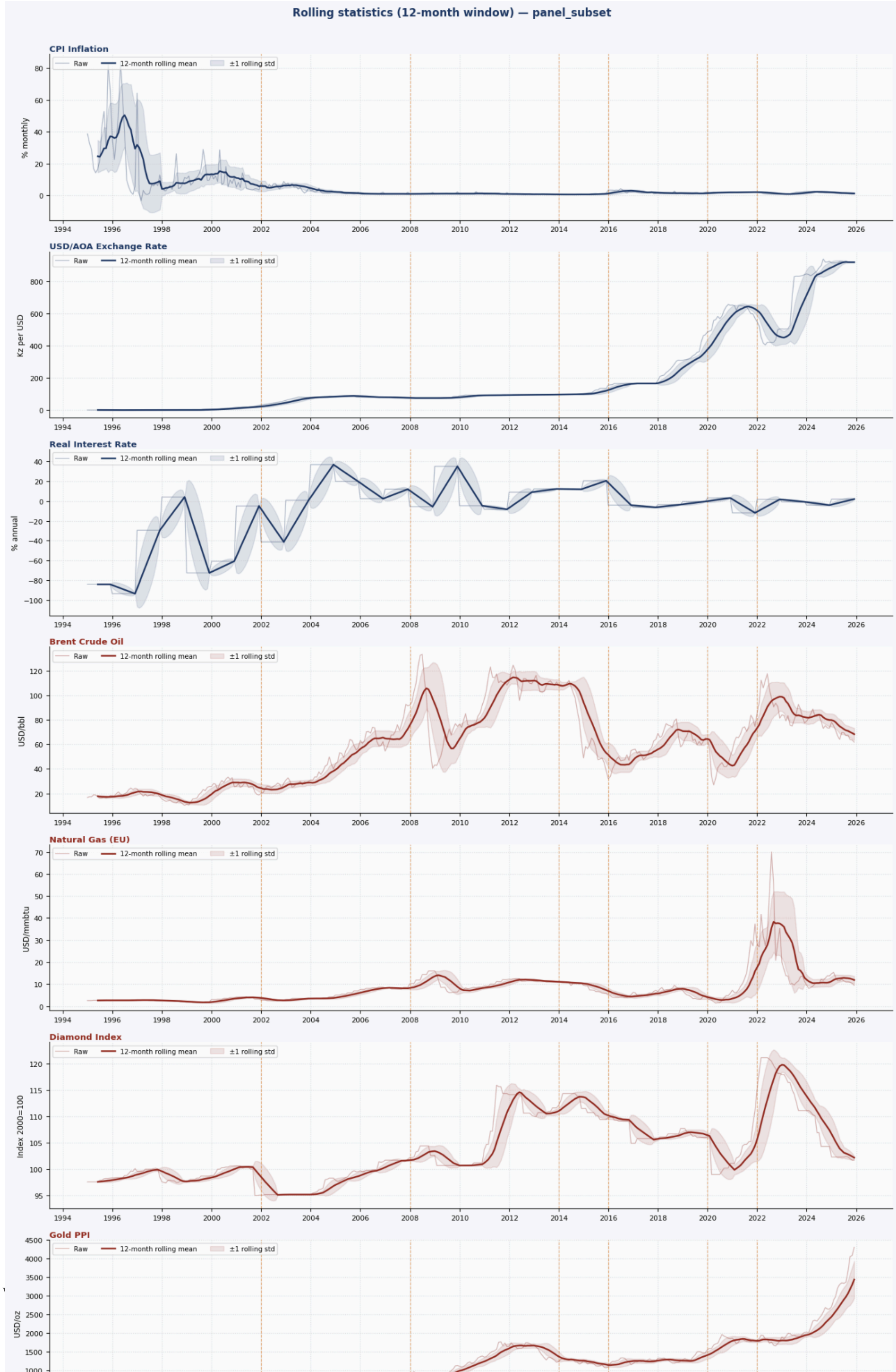


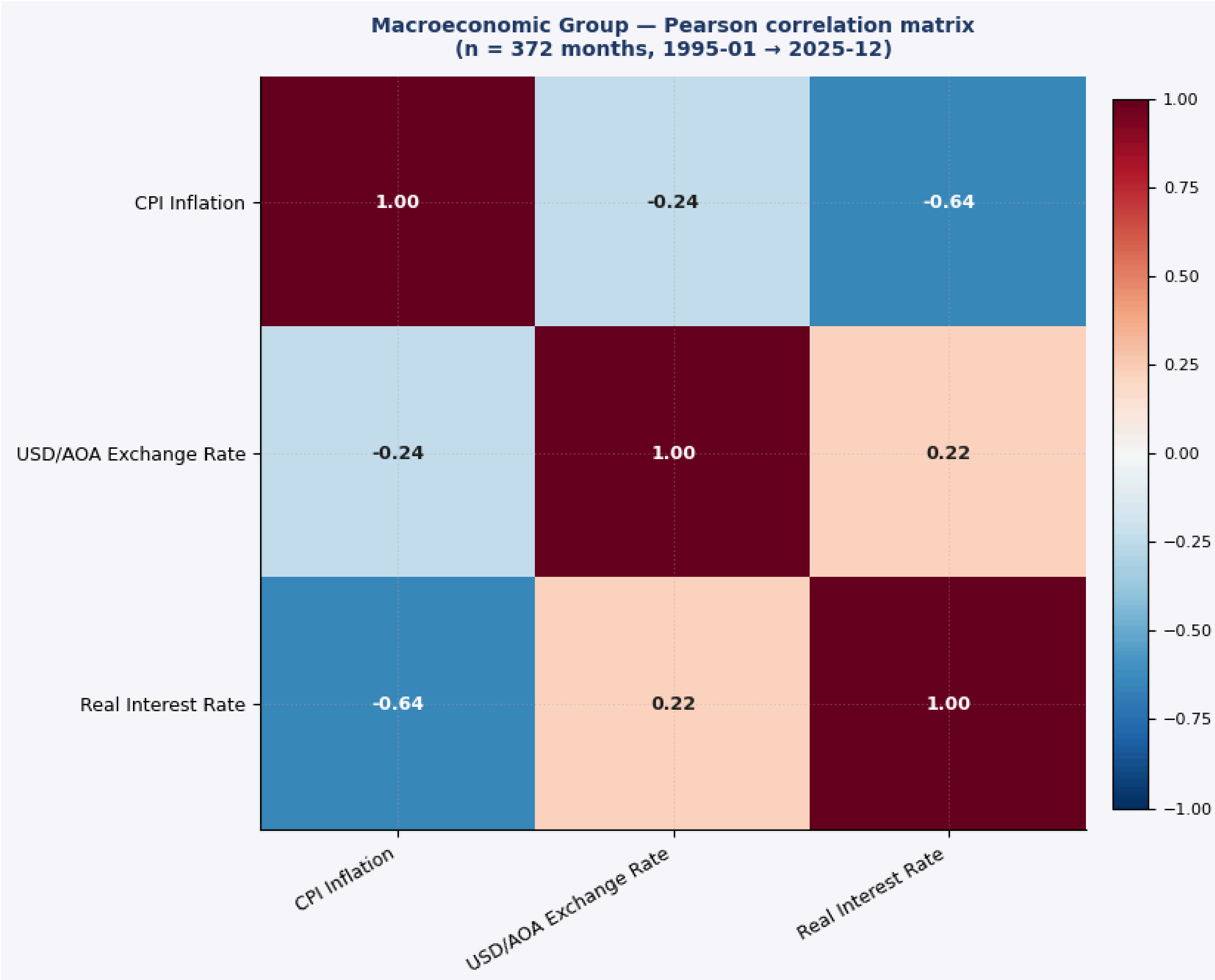


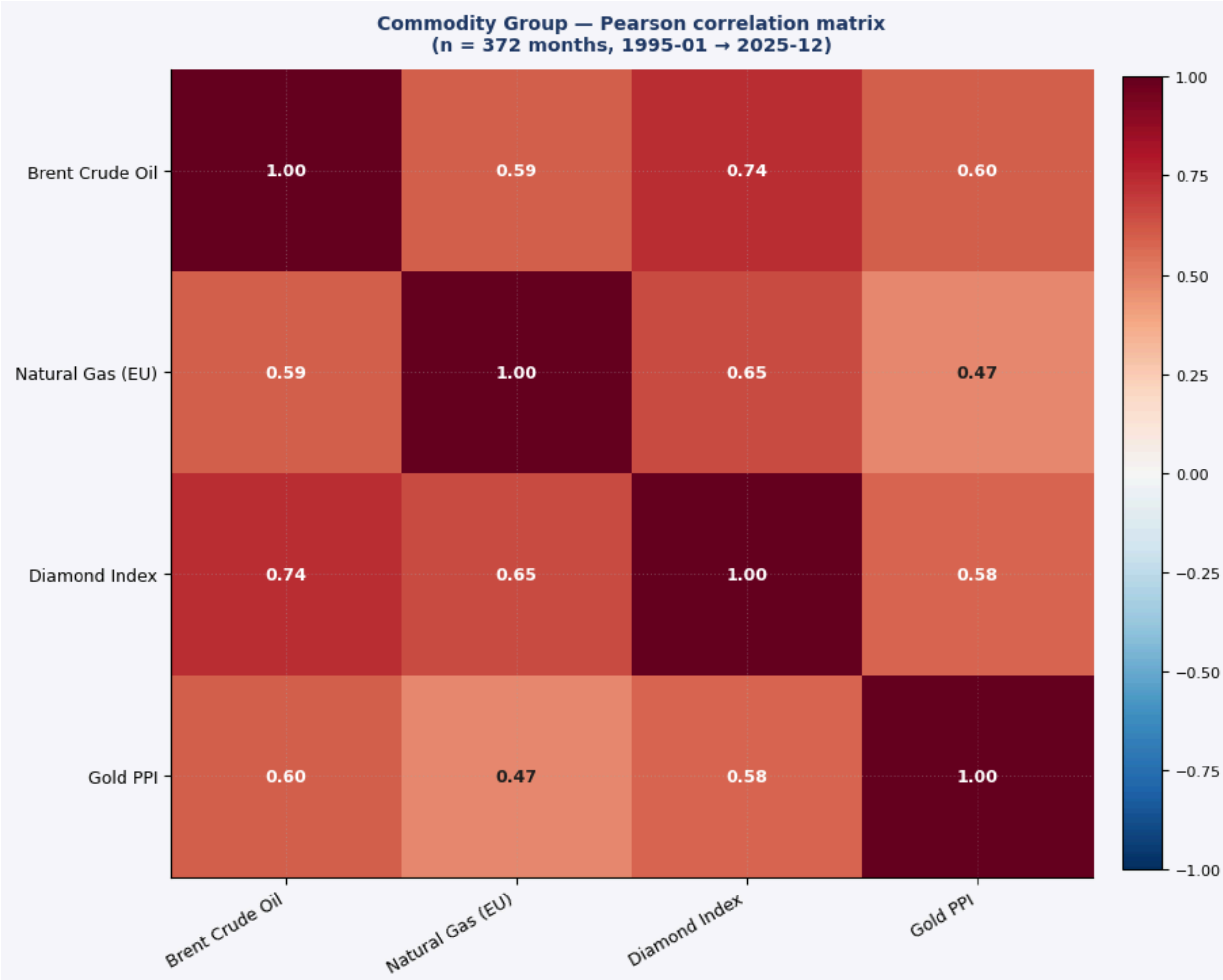

Commodity Group — panel_subset time series
Brent Crude Oil [n = 372 | 1995-01 → 2025-12]
USD/bbl
Natural Gas (EU) [n = 372 | 1995-01 → 2025-12]
USD/mmbtu
Diamond Index [n = 372 | 1995-01 → 2025-12]
Index 2000=100
Gold PPI [n = 372 | 1995-01 → 2025-12]
USD/oz
Structural breaks: 2008 · 2014 · 2016 · 2020 · 2022

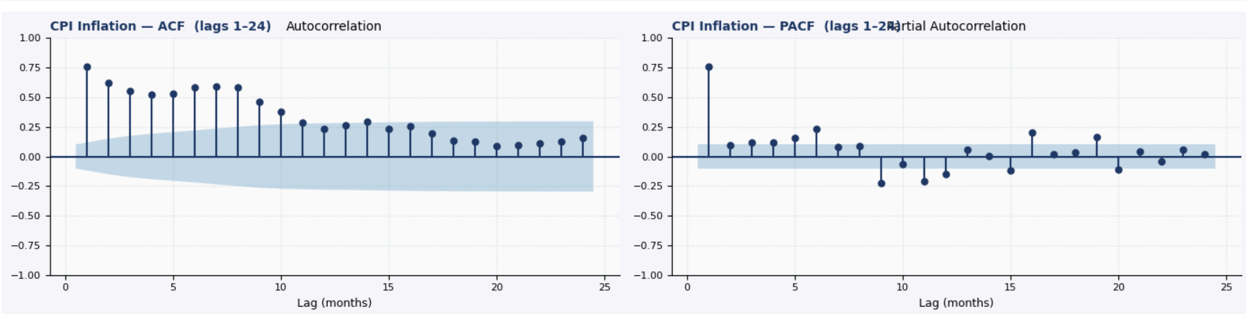

Distribution plots — panel_subset (histogram + KDE)
CPI Inflation
Skew = 4.60
Kurt = 24.28
n = 372
Mean 4.978
Median 1.595
Density
% monthly
USD/AOA Exchange Rate
Skew = 1.62
Kurt = 1.40
n = 372
Mean 203.344
Median 92.049
Density
Kz per USD
Real Interest Rate
Mean -8.614
Median -0.855
Skew = -1.36
Kurt = 1.34
n = 372
Density
% annual
Brent Crude Oil
Skew = 0.28
Kurt = -0.95
n = 372
Mean 59.041
Median 59.625
Density
USD/bbl
Natural Gas (EU)
Skew = 3.86
Kurt = 22.04
n = 372
Mean 8.006
Median 6.654
Density
USD/mmbtu
Diamond Index
Skew = 0.73
Kurt = -0.31
n = 372
Mean 103.832
Median 101.700
Density
Index 2000=100
Gold PPI
Skew = 1.14
Kurt = 1.95
n = 372
Mean 1080.997
Median 1125.903
Density
USD/oz

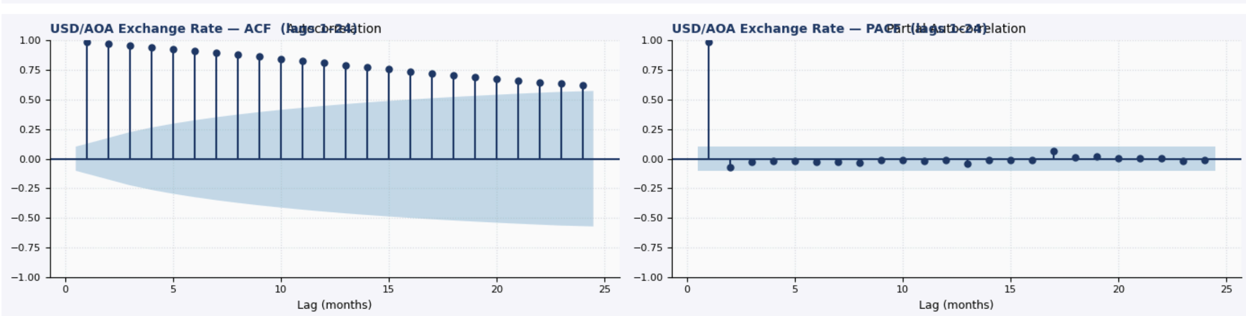

Box plots by structural break regime — panel_subset
CPI Inflation
% monthly
USD/AOA Exchange Rate
Kz per USD
Real Interest Rate
% annual
Brent Crude Oil
USD/bbl
Natural Gas (EU)
USD/mmbtu
Diamond Index
Index 2000=100
Gold PPI
USD/oz
Pre-2014
2014–2019
2020–2024
n=228
n=72
n=72

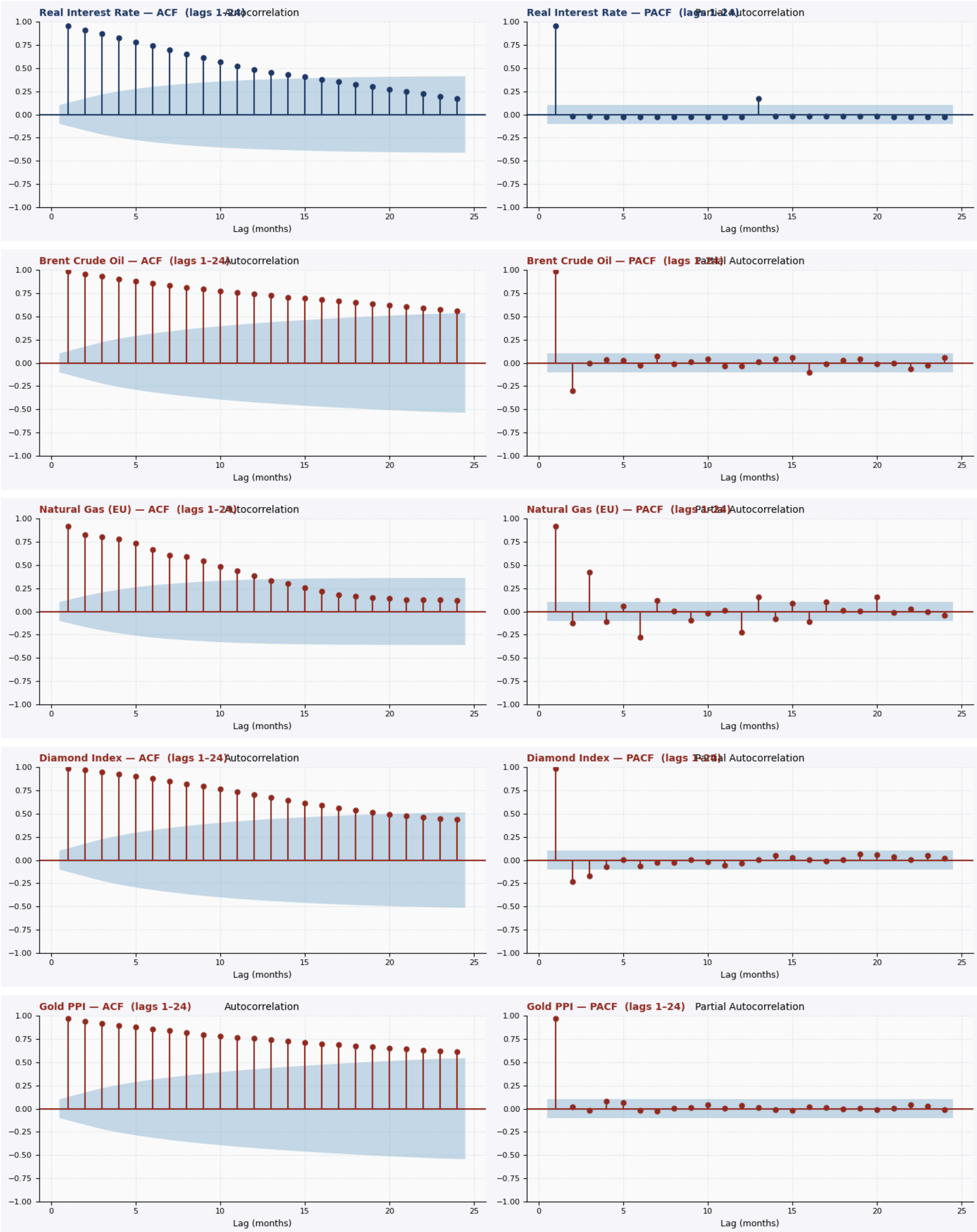

Rolling statistics (12-month window) — panel_subset
CPI Inflation
% monthly
USD/AOA Exchange Rate
Kz per USD
Real Interest Rate
% annual
Brent Crude Oil
USD/bbl
Natural Gas (EU)
USD/mmbtu
Diamond Index
Index 2000=100
Gold PPI
USD/oz
Raw
12-month rolling mean
±1 rolling std


Macroeconomic Group — Pearson correlation matrix
(n = 372 months, 1995-01 → 2025-12)
CPI Inflation
USD/AOA Exchange Rate
Real Interest Rate
1.00
-0.24
-0.64
-0.24
1.00
0.22
-0.64
0.22
1.00
CPI Inflation
USD/AOA Exchange Rate
Real Interest Rate
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Commodity Group — Pearson correlation matrix
(n = 372 months, 1995-01 → 2025-12)
Brent Crude Oil
Natural Gas (EU)
Diamond Index
Gold PPI
1.00
0.59
0.74
0.60
0.59
1.00
0.65
0.47
0.74
0.65
1.00
0.58
0.60
0.47
0.58
1.00


CPI Inflation — ACF (lags 1-24)
Autocorrelation
CPI Inflation — PACF (lags 1-24)
Partial Autocorrelation
Lag (months)
Lag (months)


USD/AOA Exchange Rate — ACF (lags 1-24)
USD/AOA Exchange Rate — PACF (lags 1-24)
Lag (months)
Lag (months)


Real Interest Rate — ACF (lags 1–24)
Autocorrelation
Real Interest Rate — PACF (lags 1–24)
Partial Autocorrelation
Brent Crude Oil — ACF (lags 1–24)
Brent Crude Oil — PACF (lags 1–24)
Natural Gas (EU) — ACF (lags 1–24)
Natural Gas (EU) — PACF (lags 1–24)
Diamond Index — ACF (lags 1–24)
Diamond Index — PACF (lags 1–24)
Gold PPI — ACF (lags 1–24)
Gold PPI — PACF (lags 1–24)
Lag (months)